

IC 121: Mechanics of Particles and waves

Course Code	IC 121
Course Name	Mechanics of Particles and waves
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech 1st year
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum

Course Contents

Part I – Classical mechanics

- **Vectors and vector calculus:** gradient, divergence and curl, line, surface and volume integrals - Helmholtz theorem. Gauss divergence, Stokes theorem - Generalized coordinates, Jacobian, Cartesian, cylindrical, and spherical coordinates. Introduction to Cartesian tensors. Vectors and vector spaces. [5 Lectures]
- **Newtonian mechanics conservation laws:** linear, angular momentum, energy- single and many particle systems. [3 Lectures]
- Oscillations as application of Newtonian mechanics, Driven damped and forced oscillations, generalized vector spaces, Fourier expansion and oscillations under periodic forces, coupled oscillations and normal modes. Nonlinear oscillations. LC circuit, simple pendulum, coupled pendulum. [4 Lectures]
- Potentials and fields, Fundamental interactions in nature. Gravitational and electrostatic potentials by point particles and extended objects. Multi-pole expansion. Poisson and Laplace equation in electrostatics. [4 Lectures]
- Constraints and generalized coordinates - Lagrangian- Lagrange's equation of motion - relation to Newtonian mechanics - Two body problem - type of orbitals Variational principle of mechanics. [7 Lectures]
- Legendre transform, Hamiltonian mechanics, phase space representation Introduction to many body mechanics. [5 Lectures]

Part II Introduction to Quantum Mechanics

- Inadequacy of classical mechanics, Black body radiation, photo-electric effect, Classical unstable atoms, Bohr model of hydrogen atom, Frank-Hertz experiment. [3 Lectures]
- Uncertainty principle, Phase space and Hilbert space, Postulates of quantum mechanics, Schrödinger equation, observations and measurements, principle of superposition, operators and state functions, expectation value. [5 Lectures]
- Applications of Schrödinger equation, Particle in an infinite square well potentials, Harmonic oscillator, rigid rotor and two body (Hydrogen atom) problem. [4 Lectures]

Text Books

1. S T Thorton and J B Marion, **Classical dynamics of Particles and systems**
2. Liboff, **Introductory Quantum mechanics**
3. Eisberg and Resnick, **Quantum physics of atoms, molecules**

References

1. R P Feynman, **The Feynman Lectures on Physics** Vol. I
2. C. Kittel, W D Knight, M A Ruderman, **Mechanics : Berkley Physics course I**
3. R Douglas Gregory, **Classical mechanics**
4. T W B Kibble and F H Berkshire, **Classical mechanics**
5. D Morin, **Introduction to classical mechanics with problems and solutions**
6. D J Griffiths, **Introduction to quantum mechanics**
7. D A B Miller, **Quantum mechanics for scientists and engineers**
8. C Cohen-Tannoudji, B, **Quantum mechanics**, Vol. I